

BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

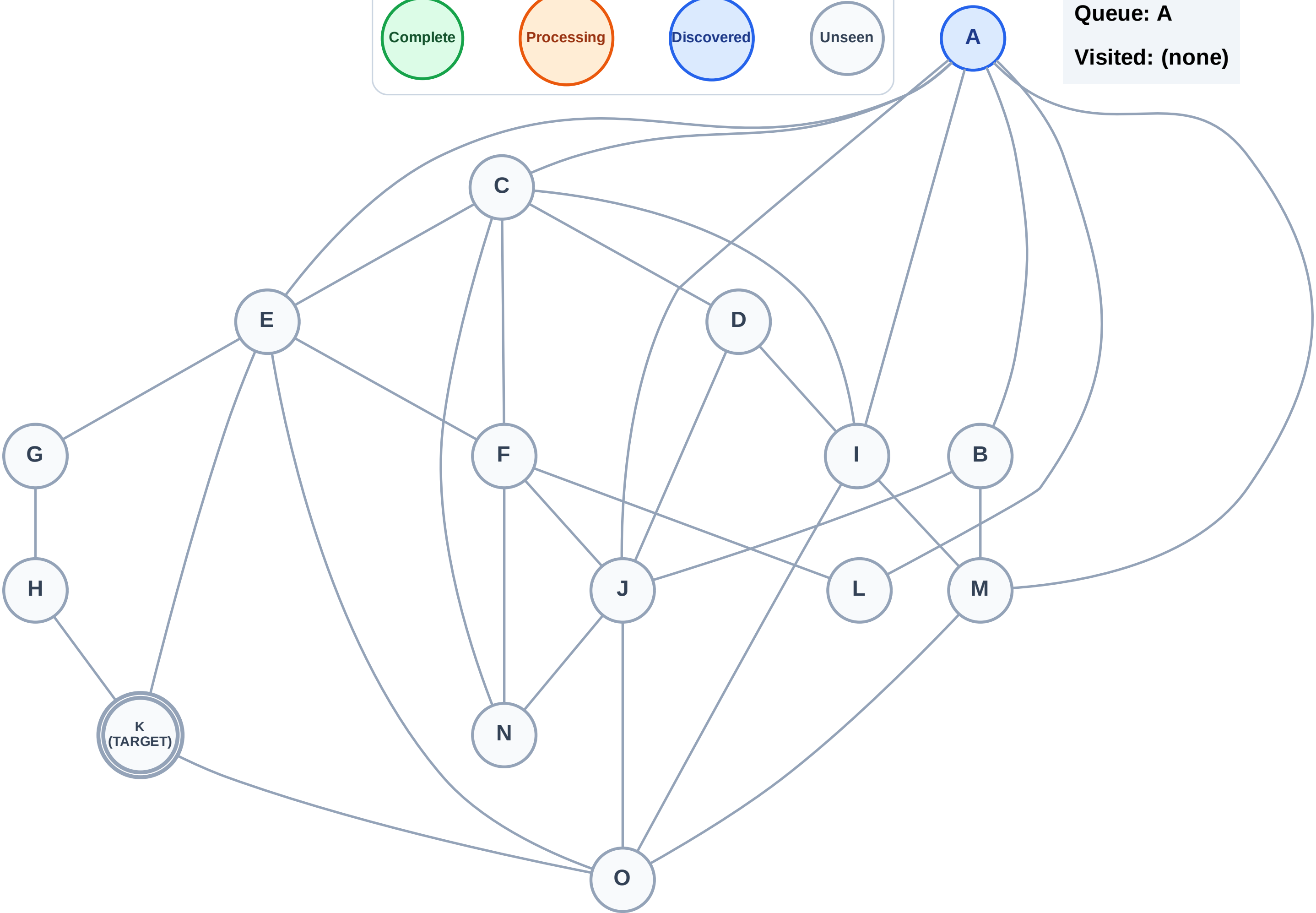
Complete

Processing

Discovered

Unseen

Queue: A
Visited: (none)



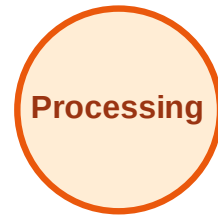
BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



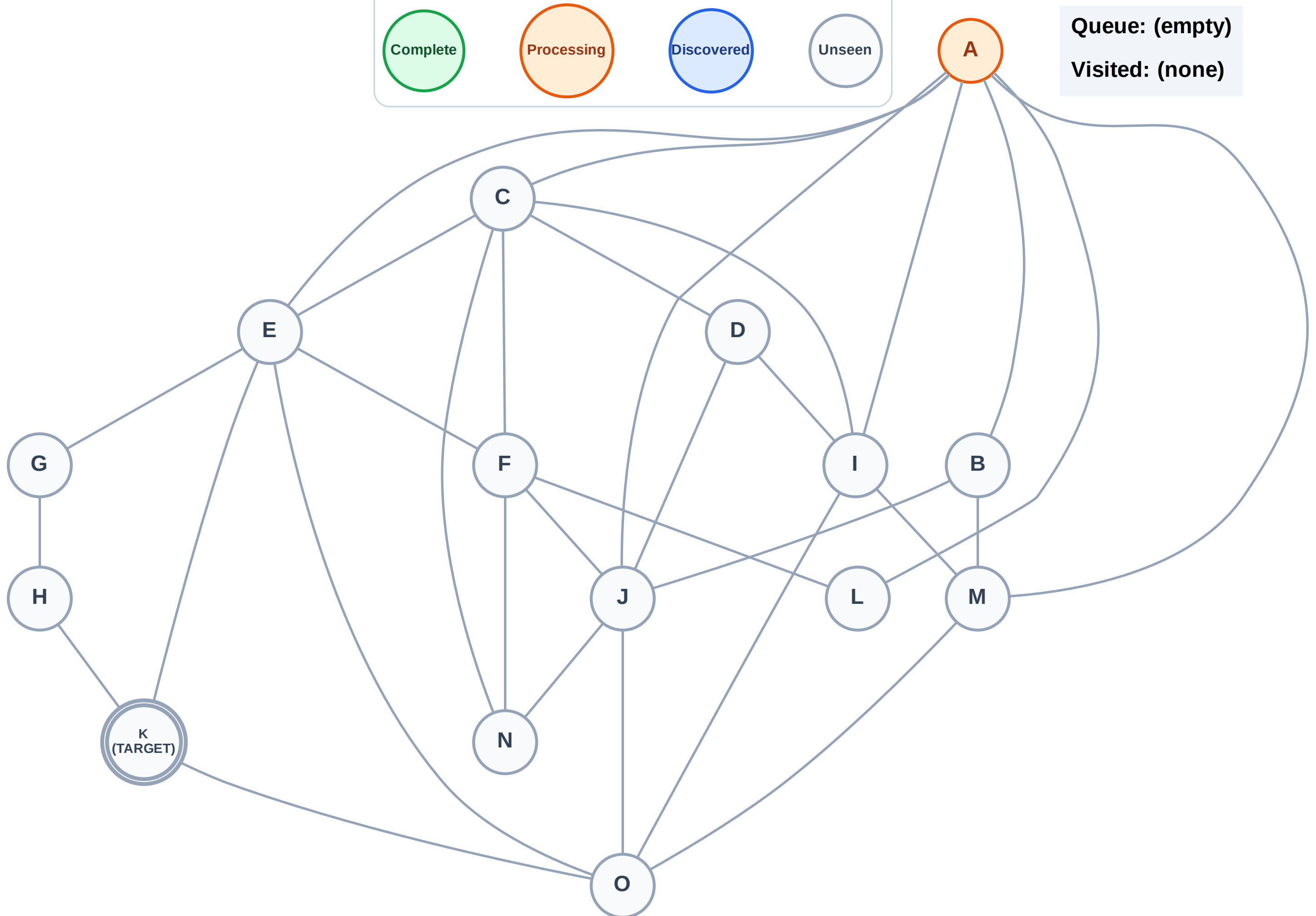
Discovered



Unseen

Queue: (empty)

Visited: (none)



BFS Traversal

Step 3: Discover B, C, E, I, J, L, M from A

Legend

Complete

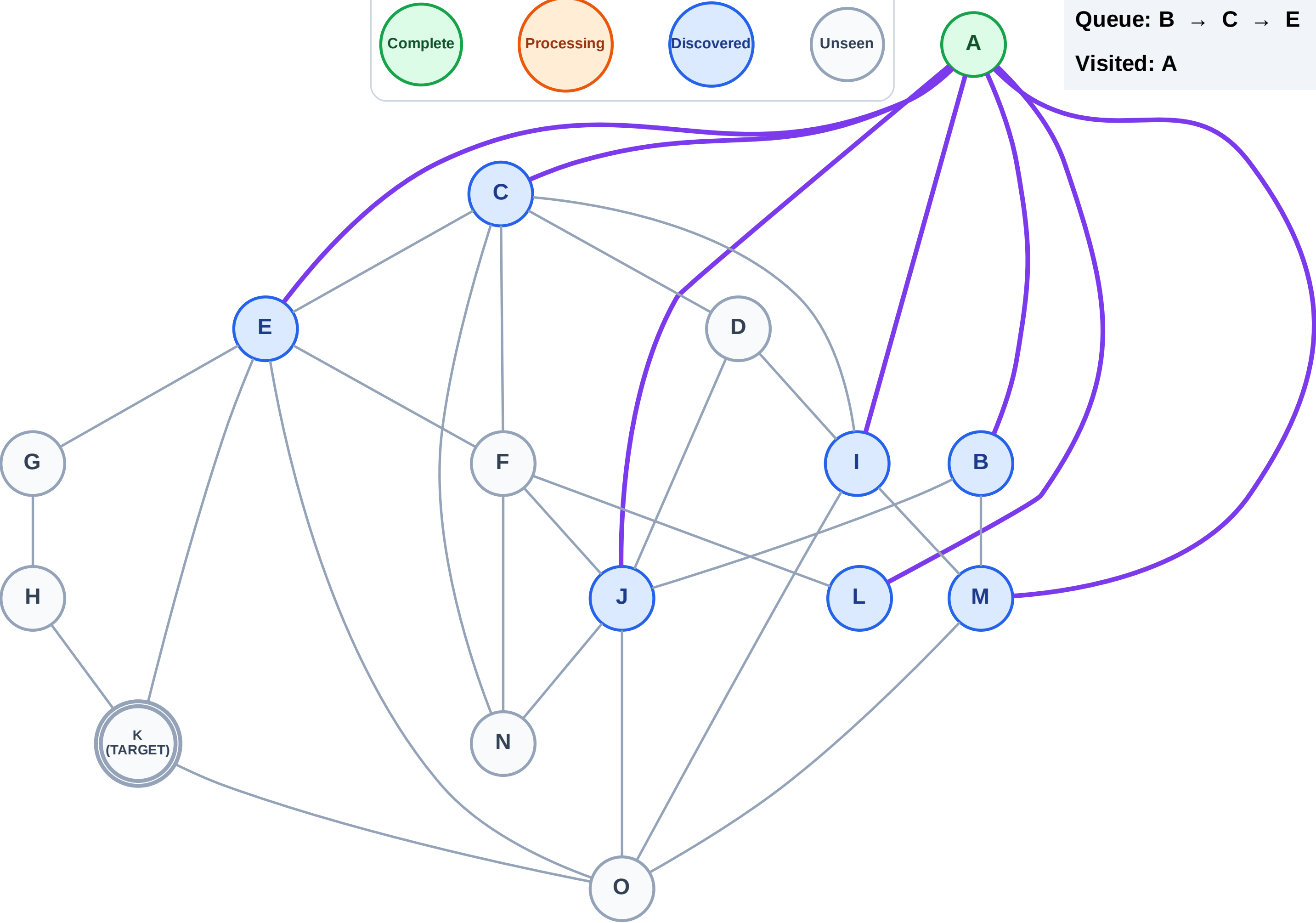
Processing

Discovered

Unseen

Queue: B → C → E → I → J → L → M

Visited: A



BFS Traversal

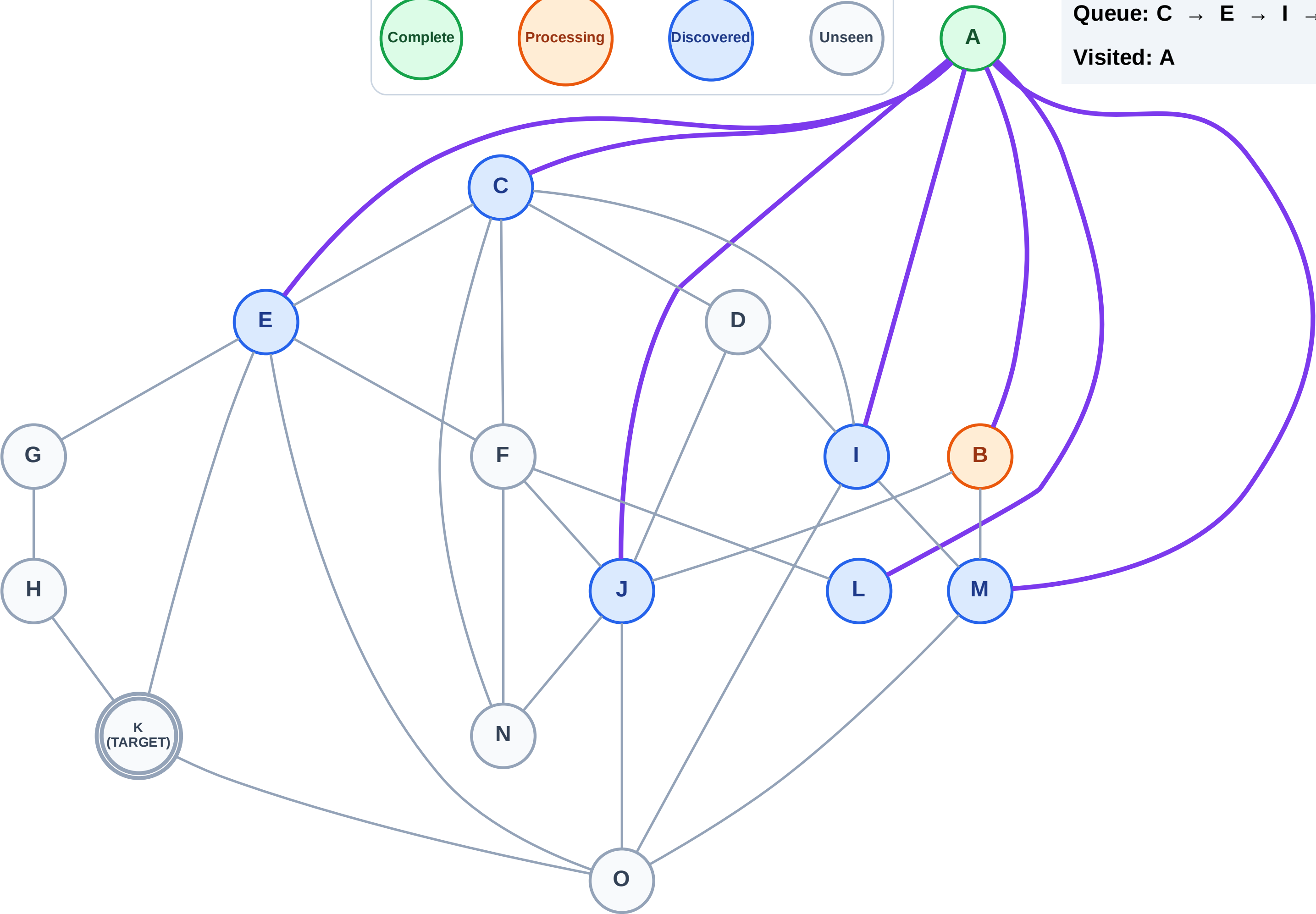
Step 4: Process node B

Legend



Queue: C → E → I → J → L → M

Visited: A



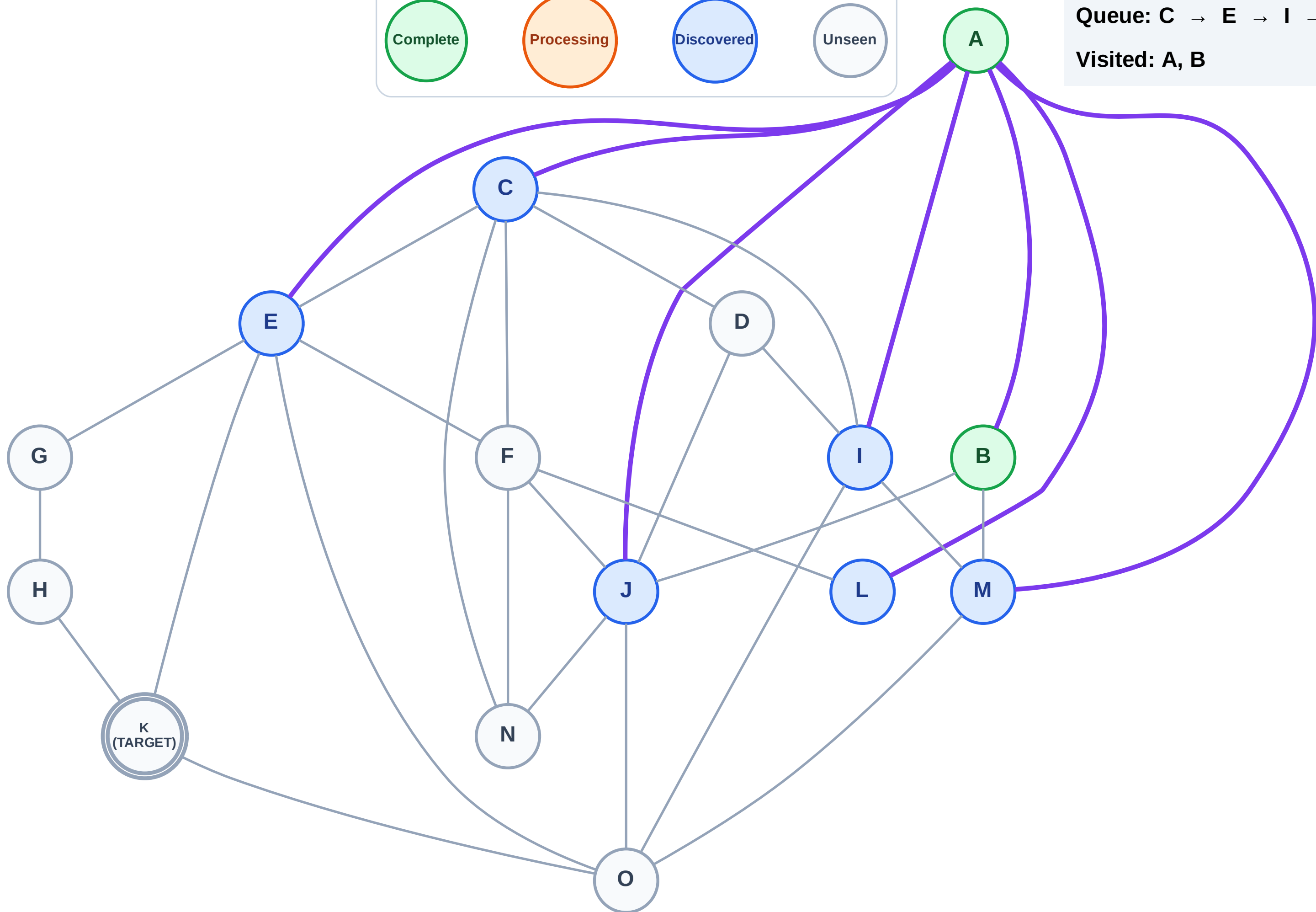
BFS Traversal

Step 5: No new neighbors from B

Legend



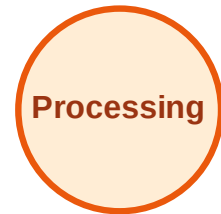
Queue: C → E → I → J → L → M
Visited: A, B



BFS Traversal

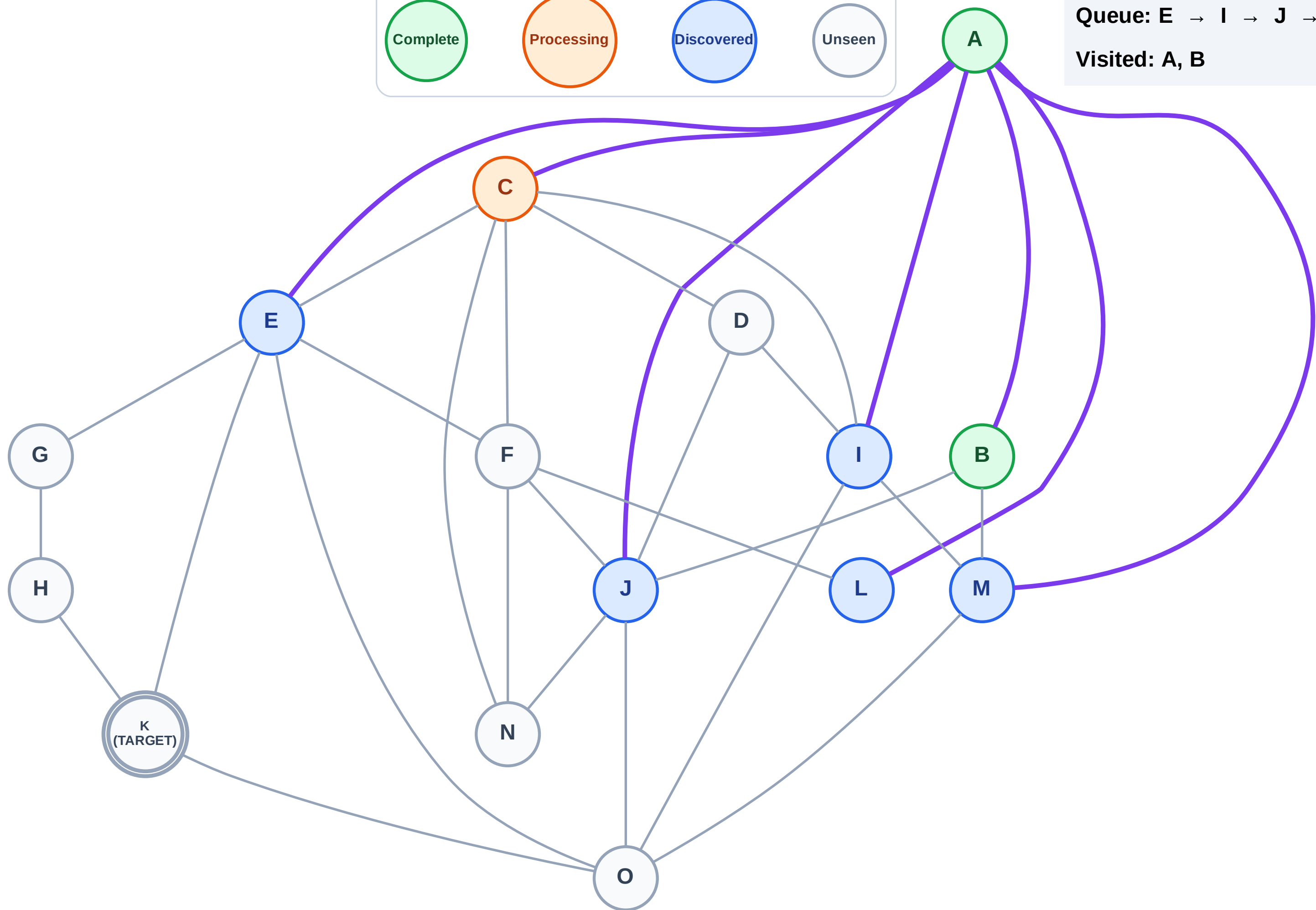
Step 6: Process node C

Legend



Queue: E → I → J → L → M

Visited: A, B



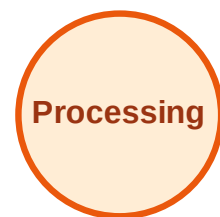
BFS Traversal

Step 7: Discover D, F, N from C

Legend



Complete



Processing

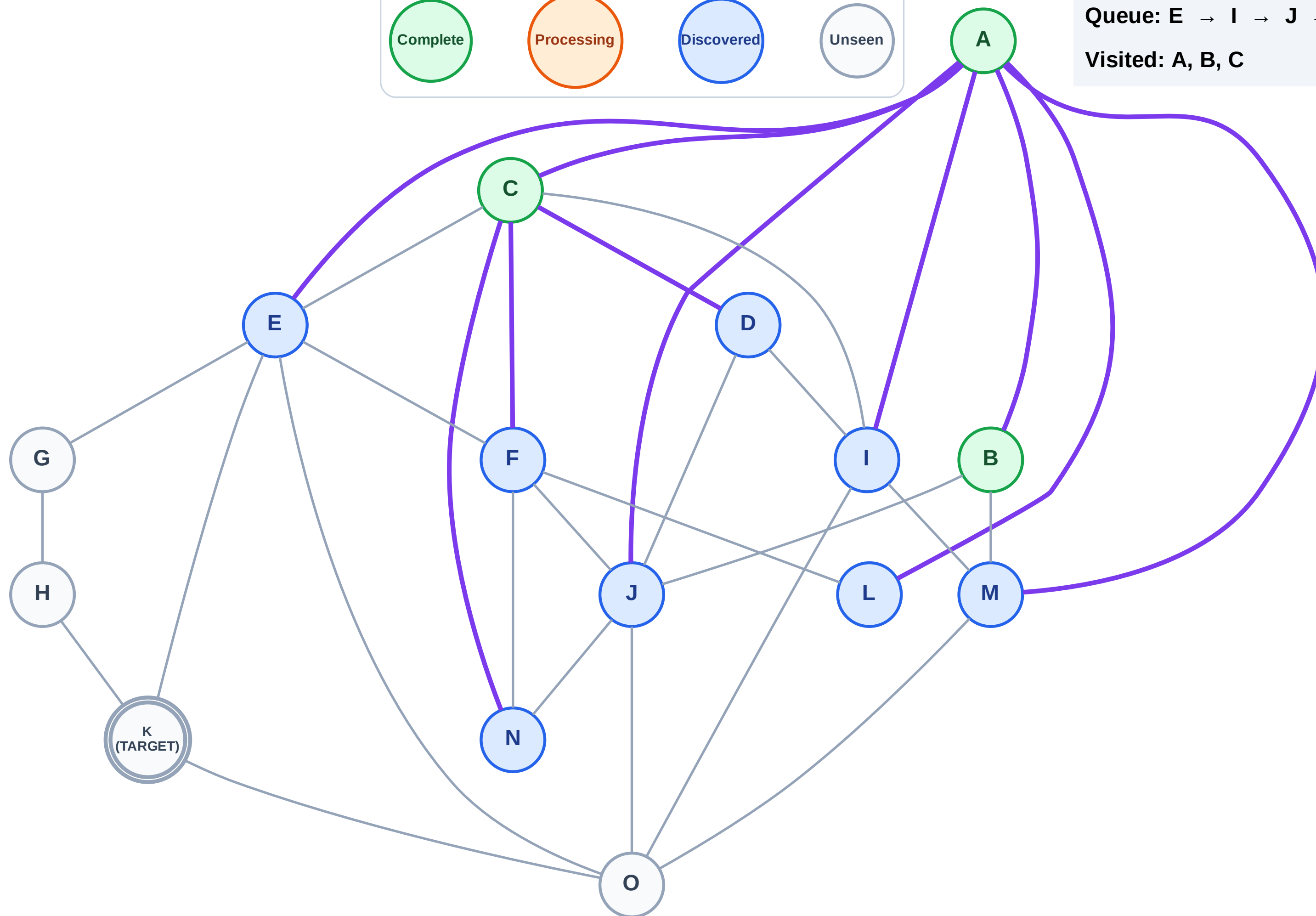


Discovered



Unseen

Queue: E → I → J → L → M → D → F → N
Visited: A, B, C



BFS Traversal

Step 8: Process node E

Legend

Complete

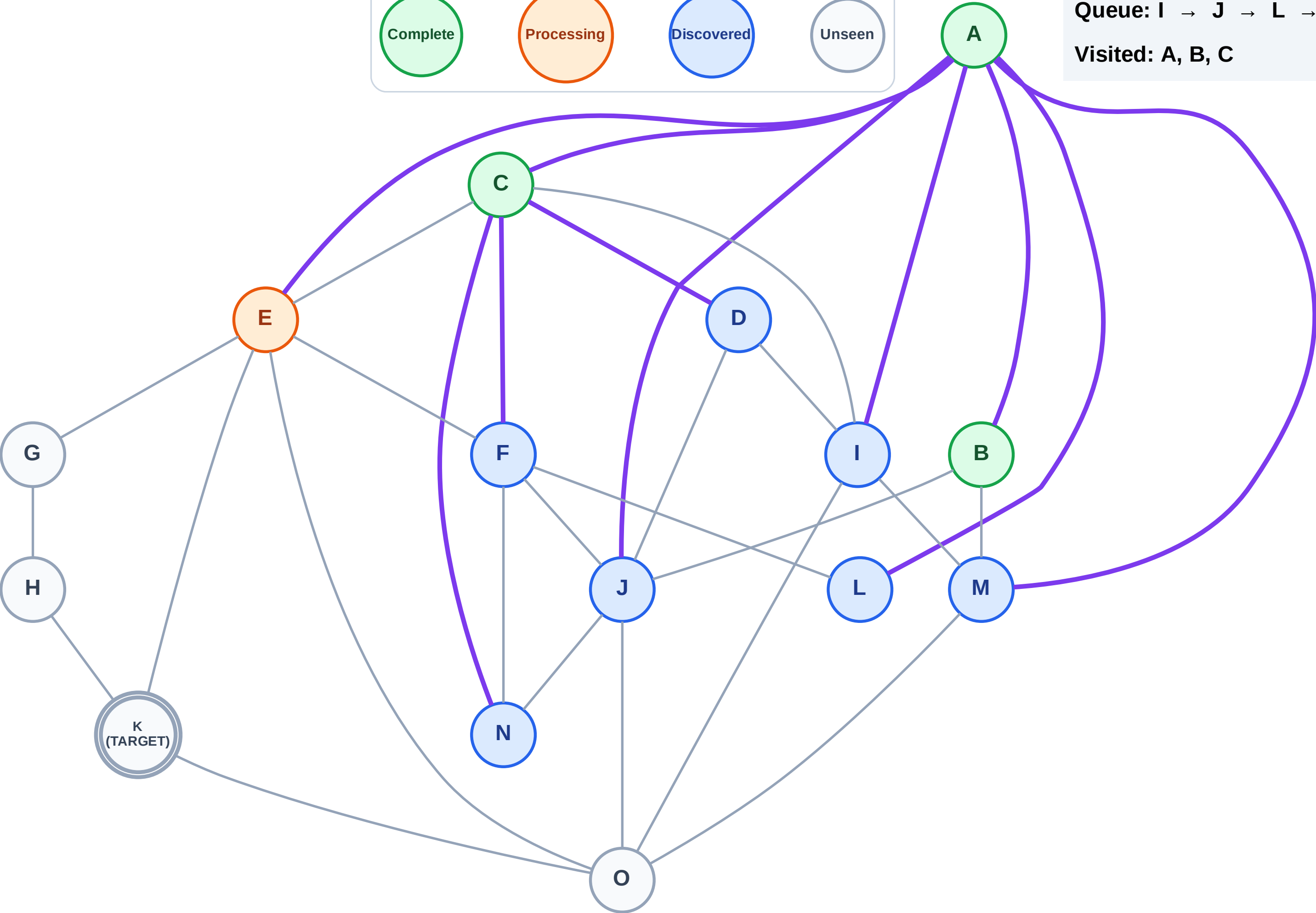
Processing

Discovered

Unseen

Queue: I → J → L → M → D → F → N

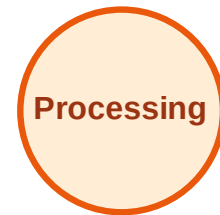
Visited: A, B, C



BFS Traversal

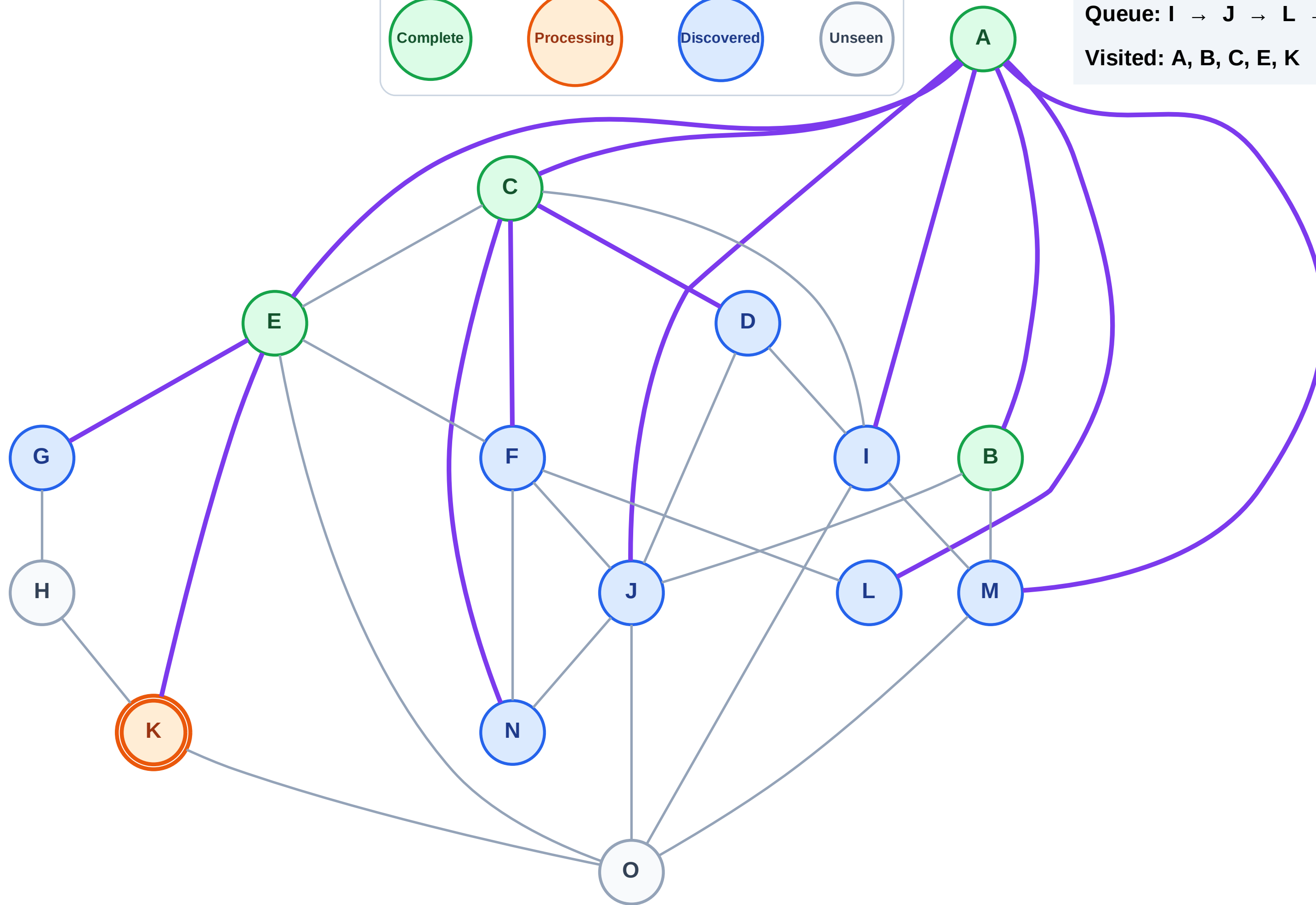
Step 9: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → E → K

Legend



Queue: I → J → L → M → D → F → N → G

Visited: A, B, C, E, K



DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

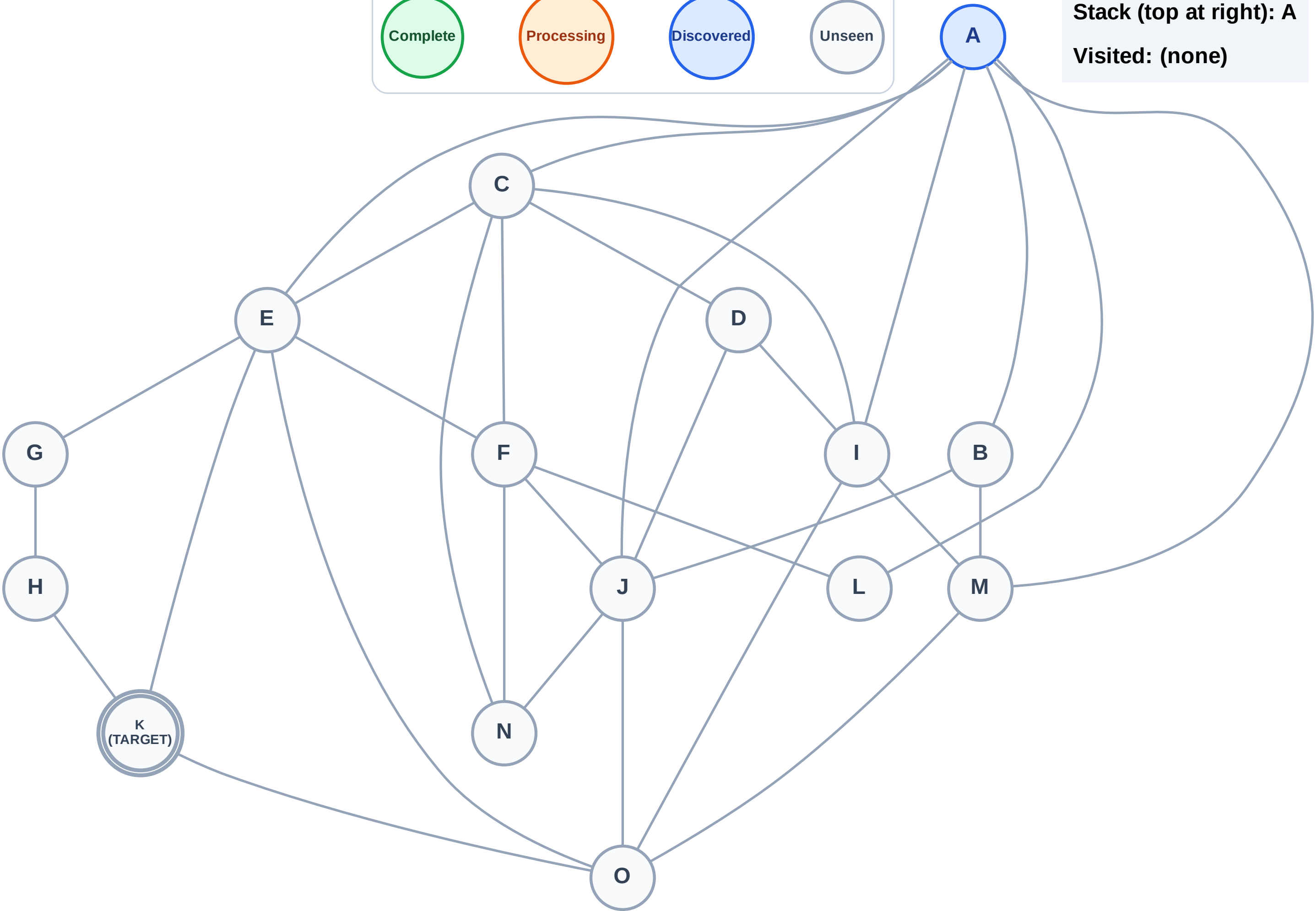
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



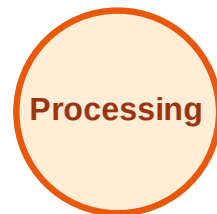
DFS Traversal

Step 2: Process node A

Legend



Complete



Processing



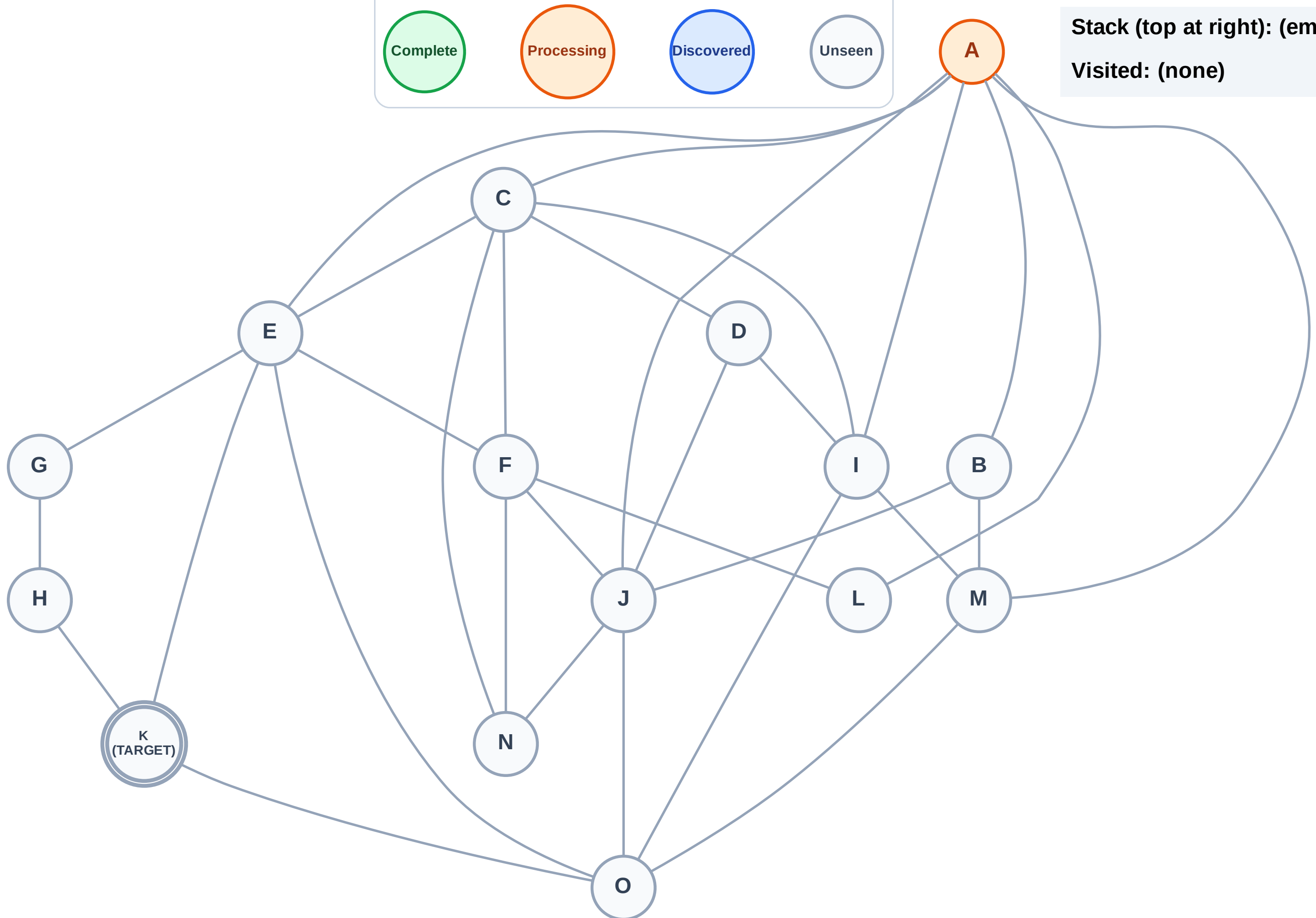
Discovered



Unseen

Stack (top at right): (empty)

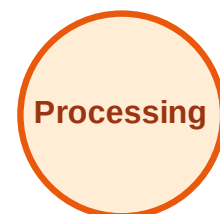
Visited: (none)



DFS Traversal

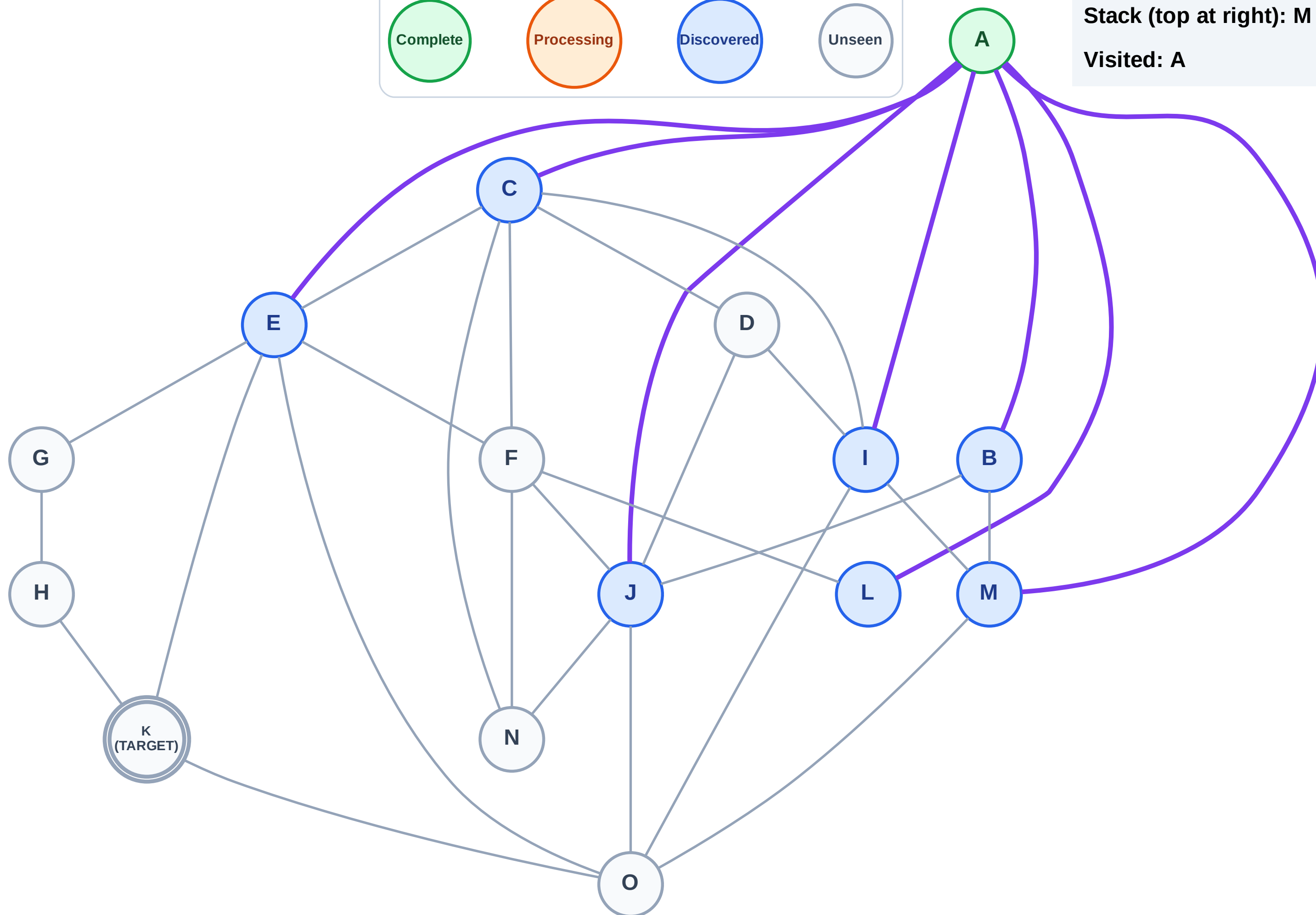
Step 3: Discover M, L, J, I, E, C, B from A

Legend



Stack (top at right): M | L | J | I | E | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

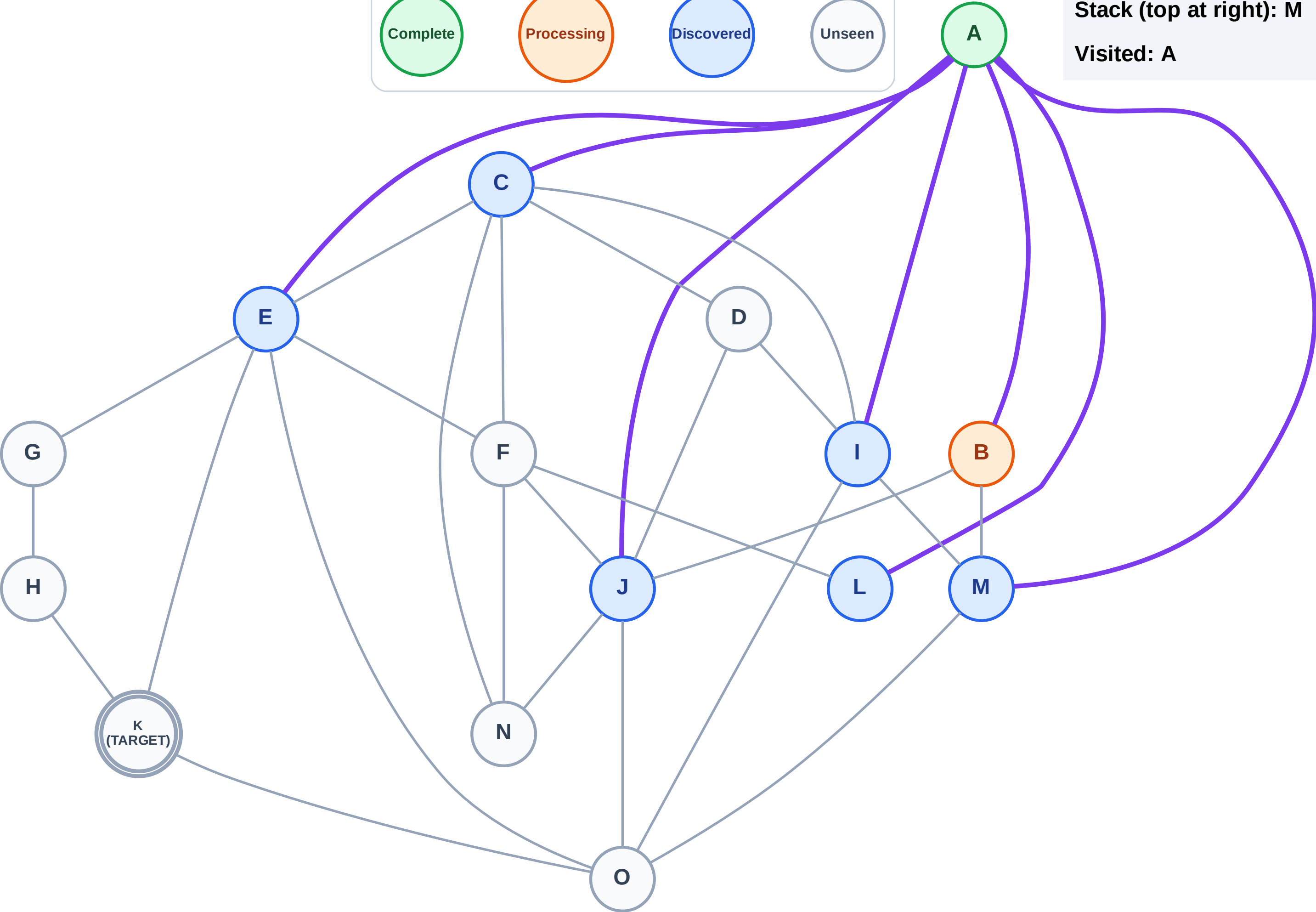
Processing

Discovered

Unseen

Stack (top at right): M | L | J | I | E | C

Visited: A



DFS Traversal

Step 5: No new neighbors from B

Legend

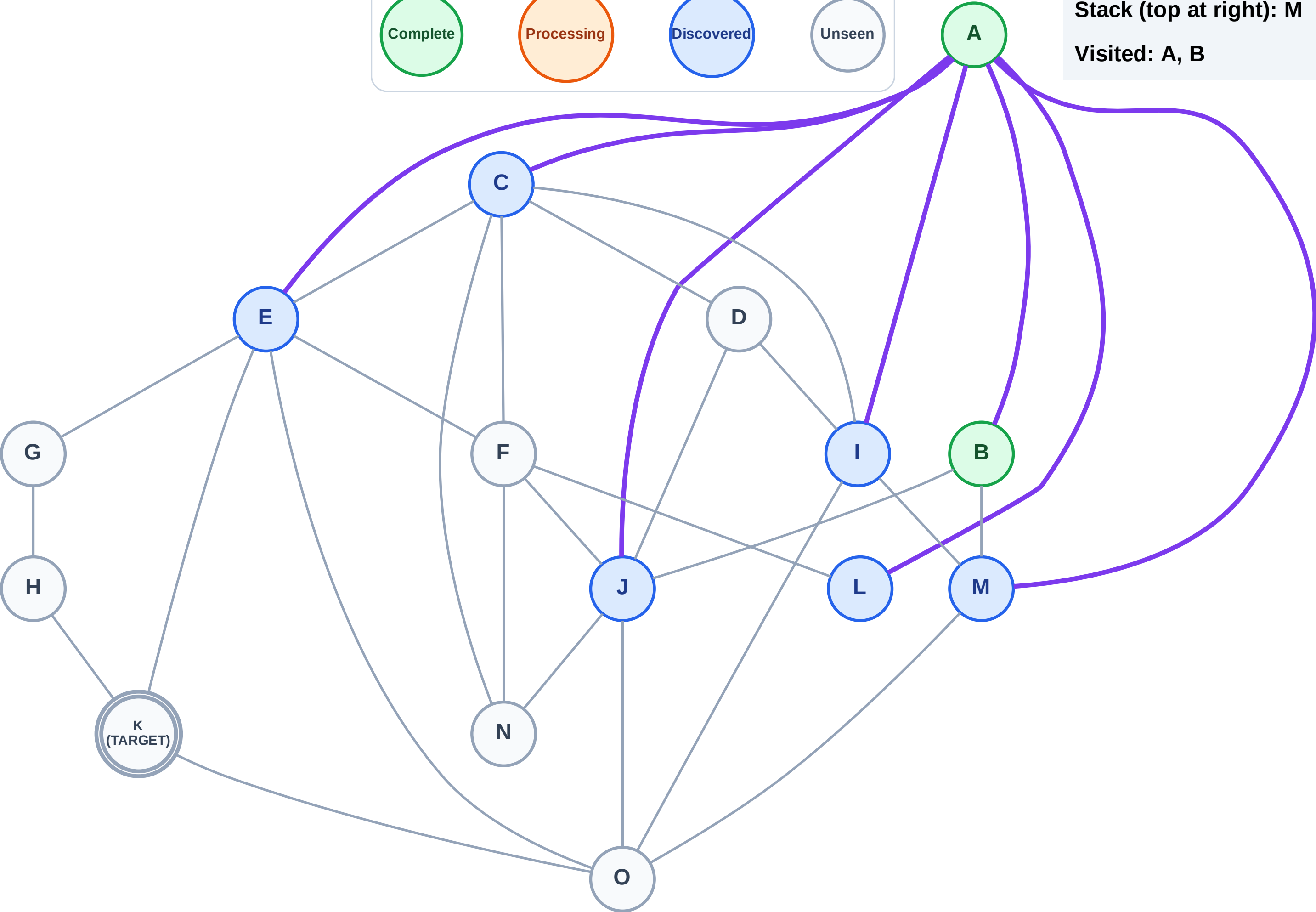
Complete

Processing

Discovered

Unseen

Stack (top at right): M | L | J | I | E | C
Visited: A, B



DFS Traversal

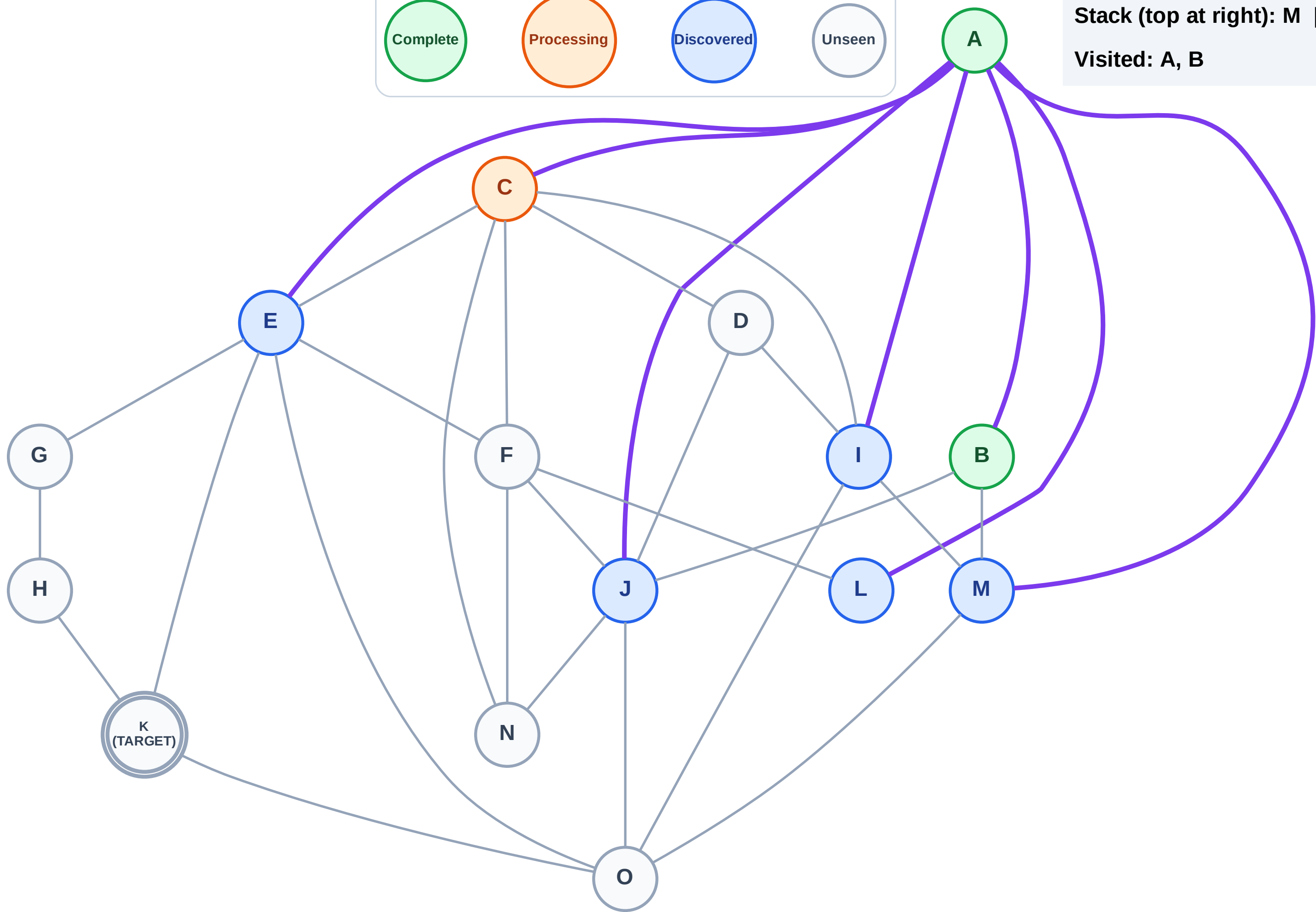
Step 6: Process node C

Legend



Stack (top at right): M | L | J | I | E

Visited: A, B



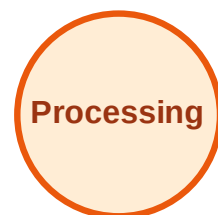
DFS Traversal

Step 7: Discover N, F, D from C

Legend



Complete



Processing



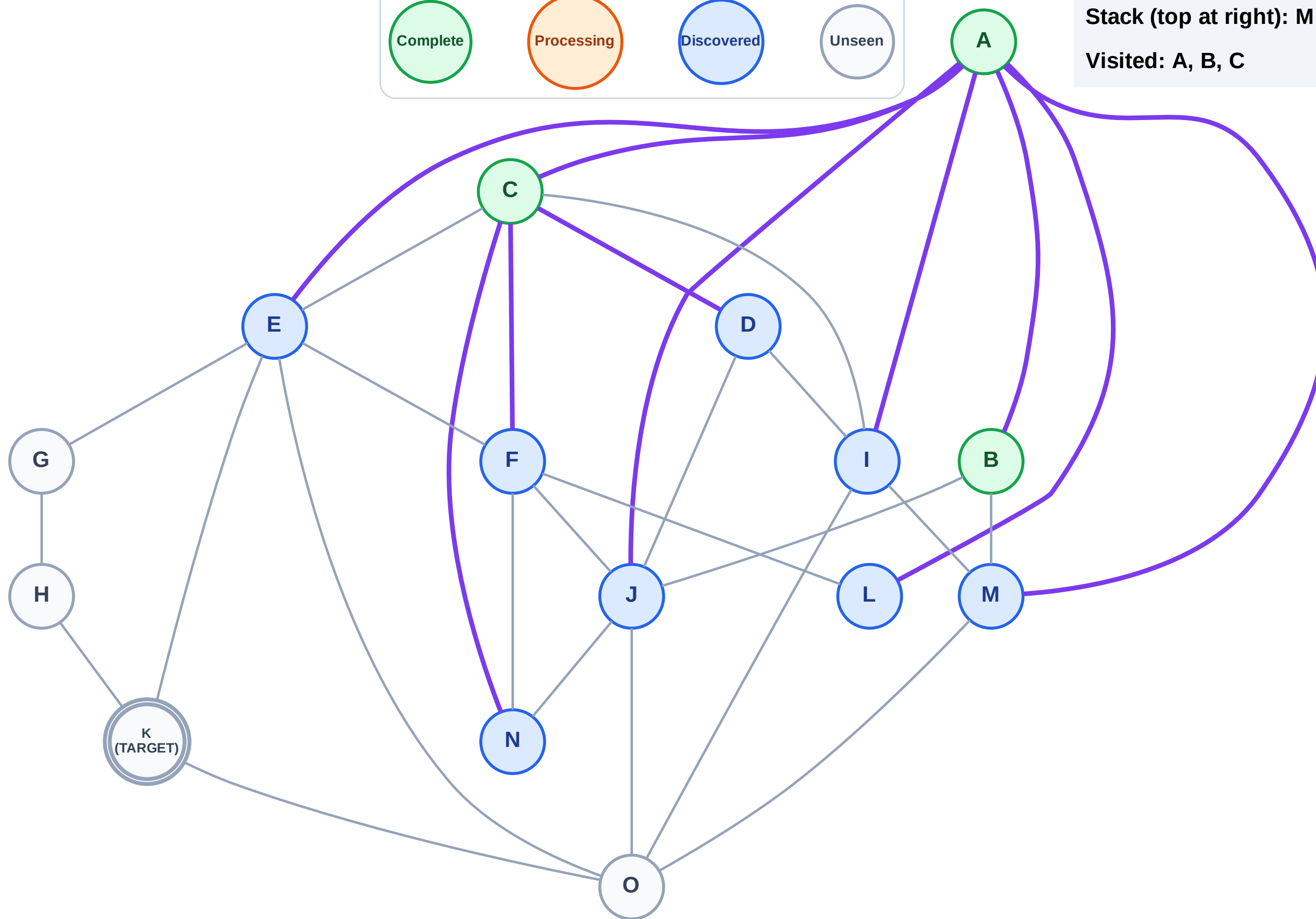
Discovered



Unseen

Stack (top at right): M | L | J | I | E | N | F | D

Visited: A, B, C



DFS Traversal

Step 8: Process node D

Legend

Complete

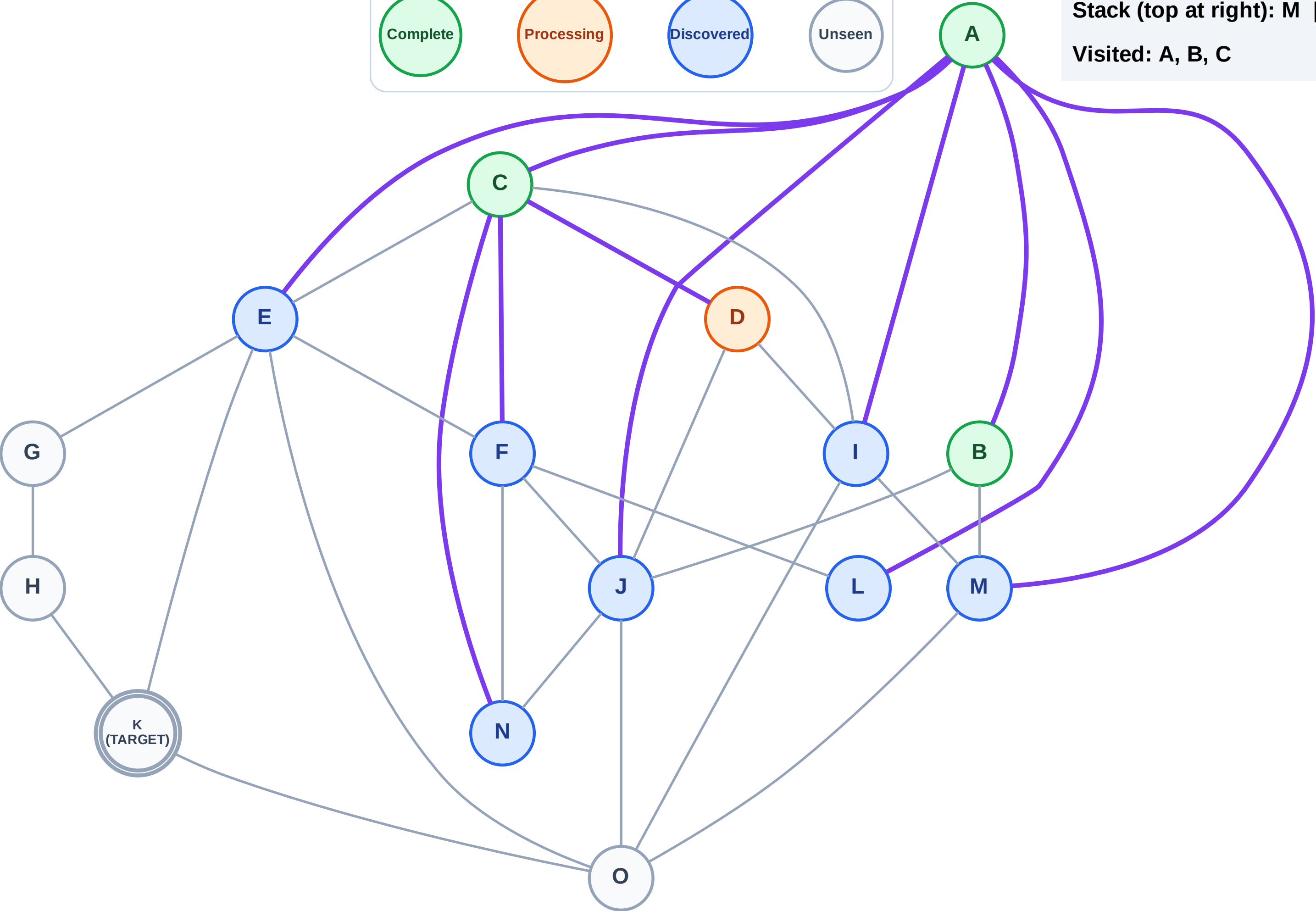
Processing

Discovered

Unseen

Stack (top at right): M | L | J | I | E | N | F

Visited: A, B, C



DFS Traversal

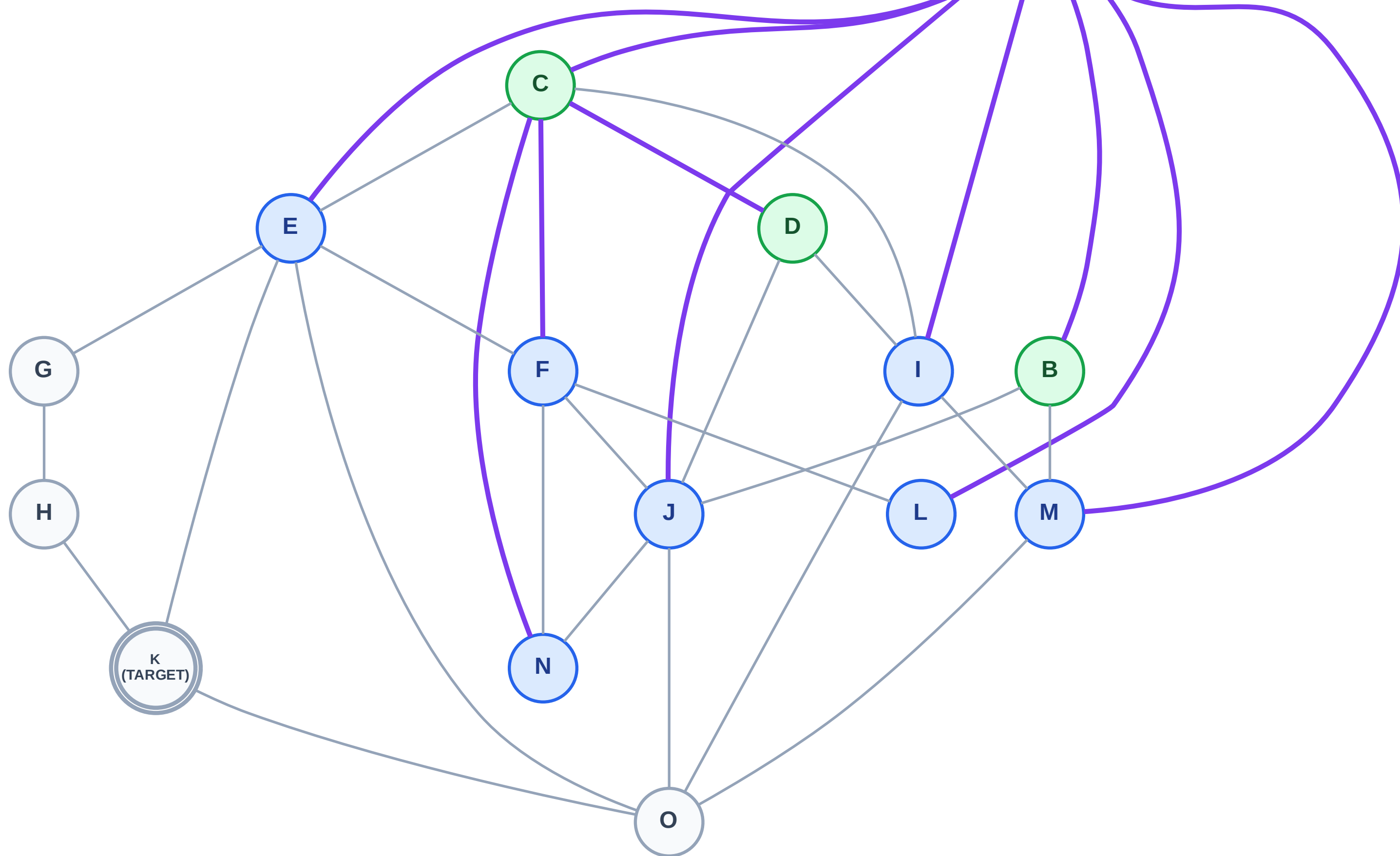
Step 9: No new neighbors from D

Legend



Stack (top at right): M | L | J | I | E | N | F

Visited: A, B, C, D



DFS Traversal

Step 10: Process node F

Legend

Complete

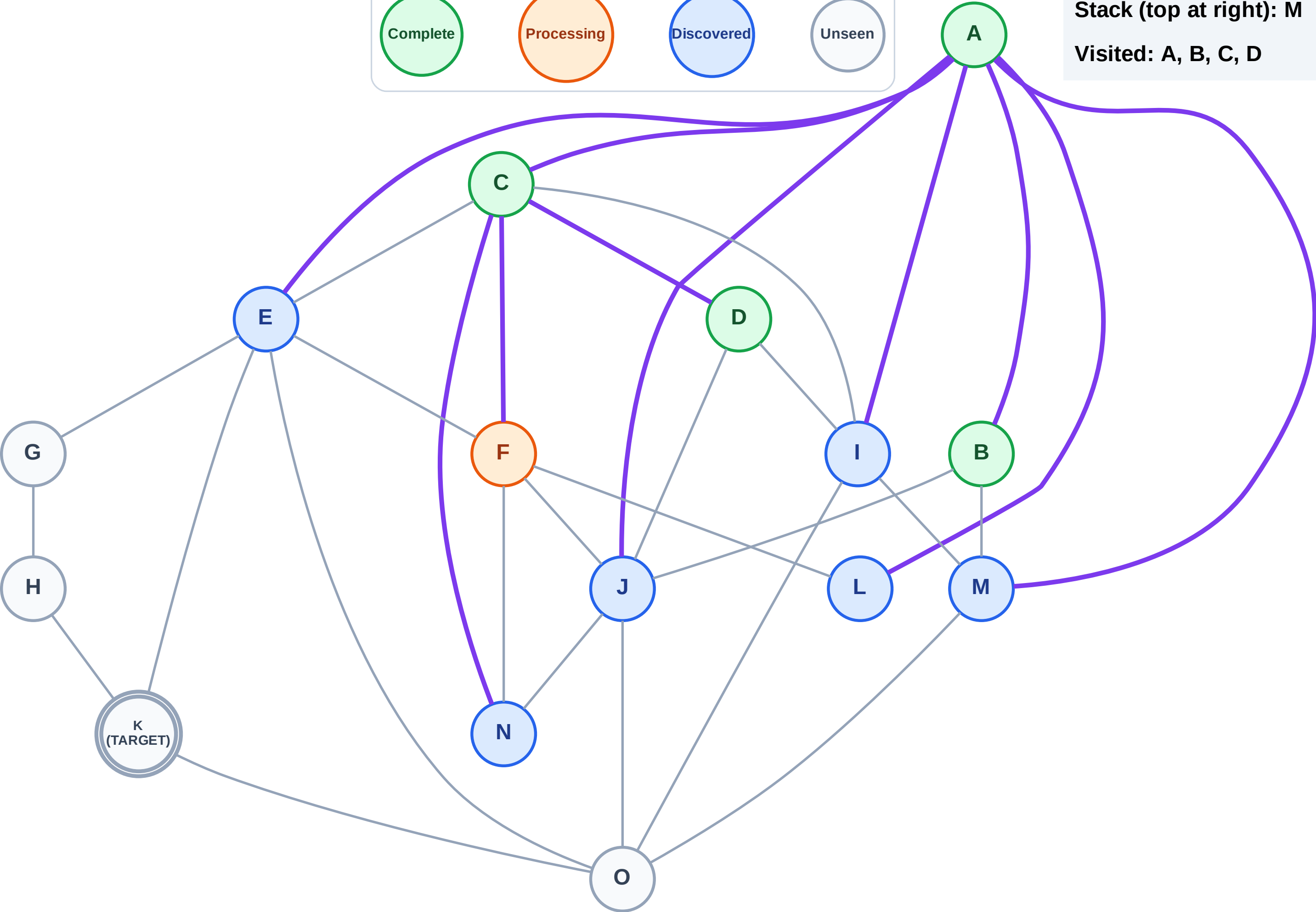
Processing

Discovered

Unseen

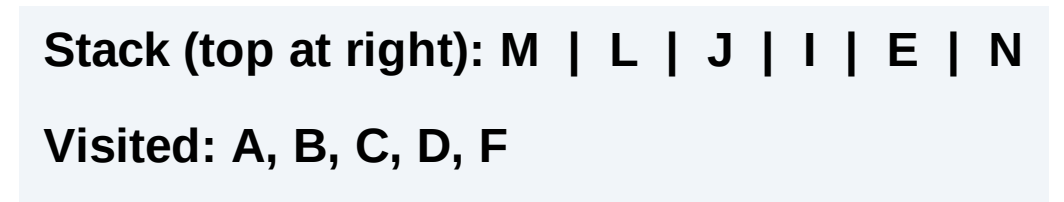
Stack (top at right): M | L | J | I | E | N

Visited: A, B, C, D



Step 11: No new neighbors from F

Step 11: No new neighbors from F



DFS Traversal

Step 12: Process node N

Legend



Complete



Processing

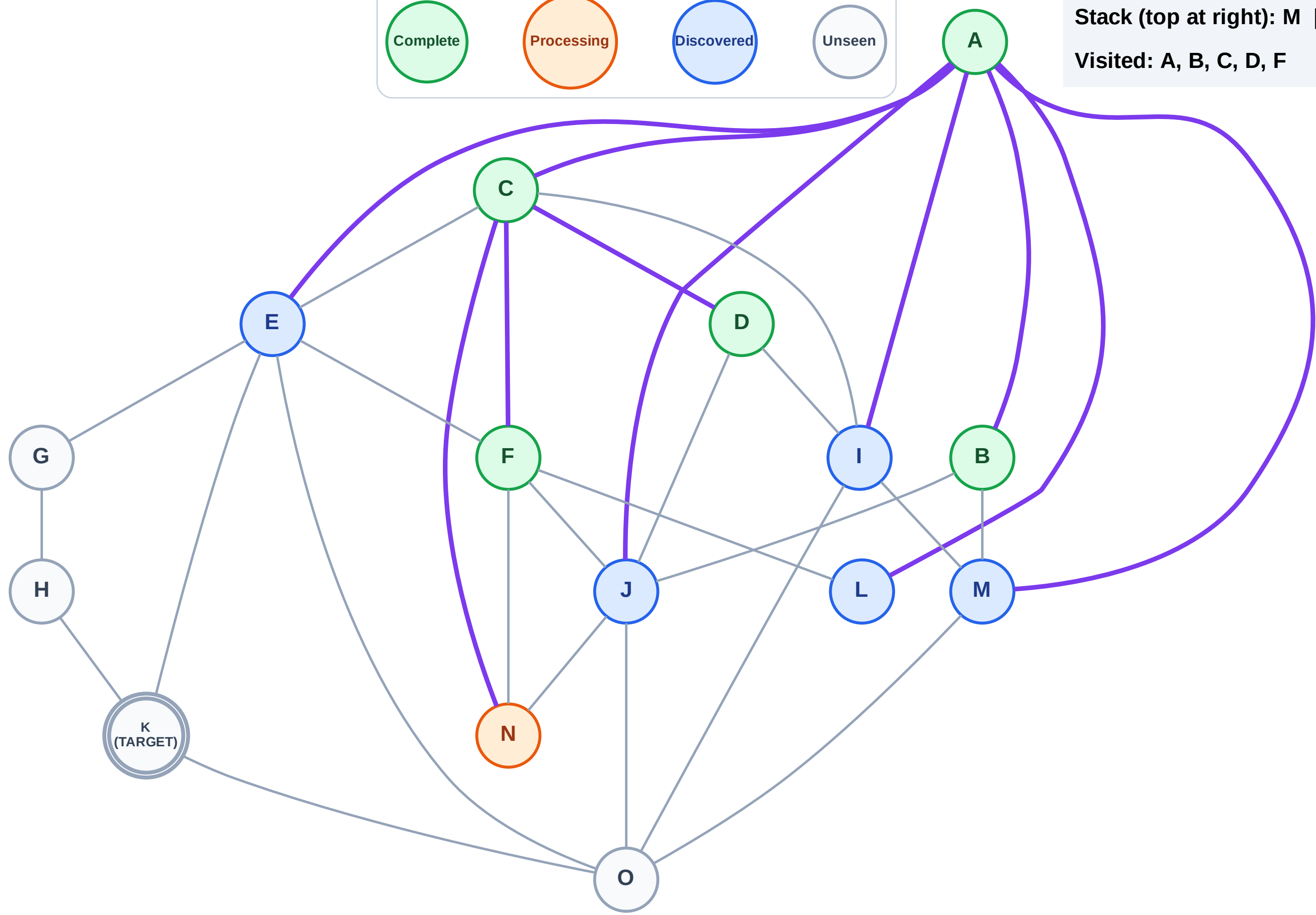


Discovered



Unseen

Stack (top at right): M | L | J | I | E
Visited: A, B, C, D, F



DFS Traversal

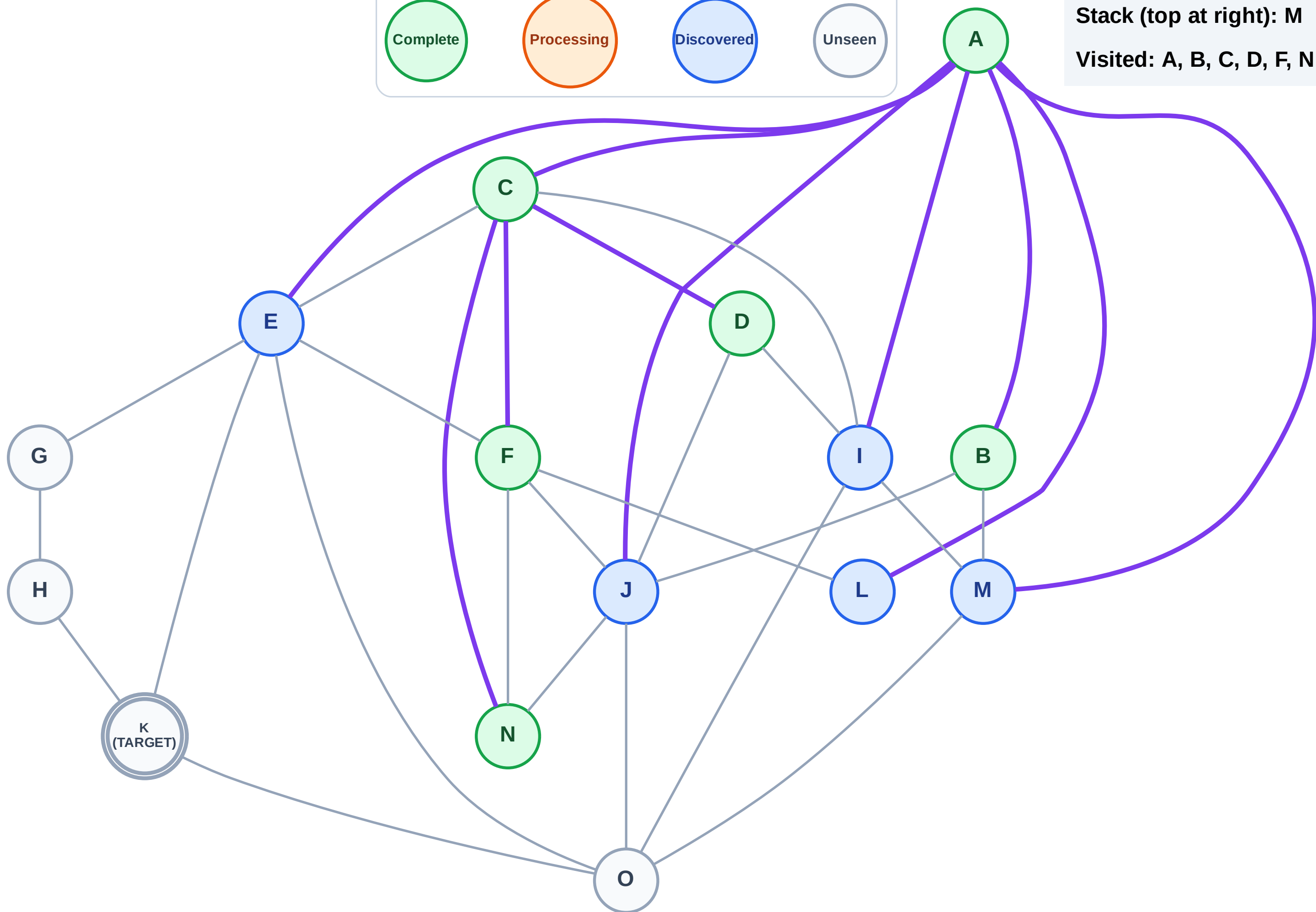
Step 13: No new neighbors from N

Legend



Stack (top at right): M | L | J | I | E

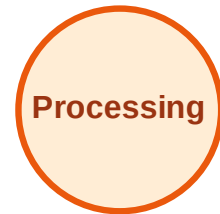
Visited: A, B, C, D, F, N



DFS Traversal

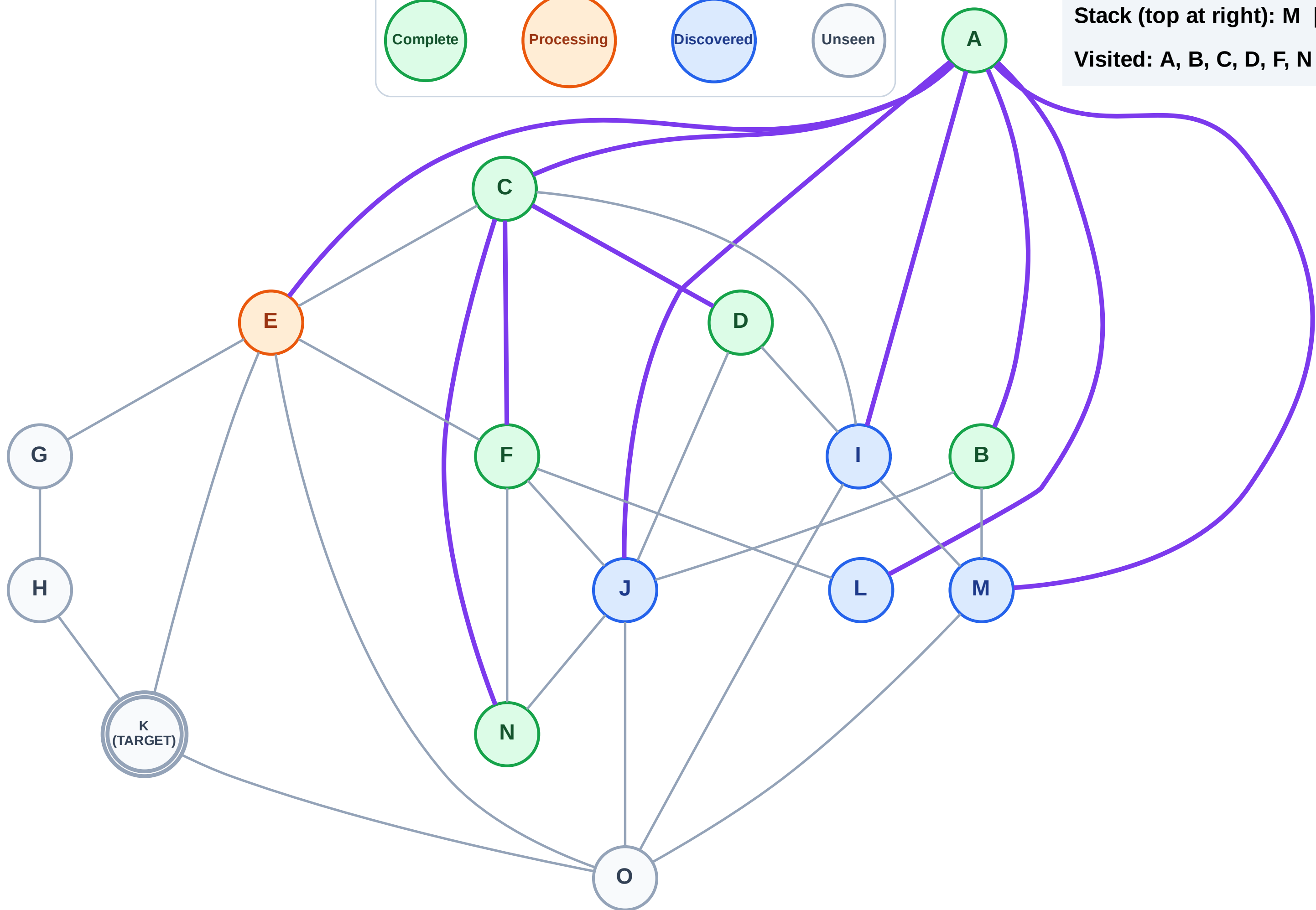
Step 14: Process node E

Legend



Stack (top at right): M | L | J | I

Visited: A, B, C, D, F, N



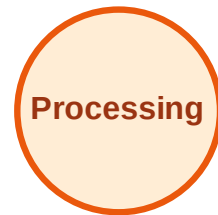
DFS Traversal

Step 15: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → E → K

Legend



Complete



Processing



Discovered



Unseen

Stack (top at right): M | L | J | I | O

Visited: A, B, C, D, E, F, K, N

